□□□□□□□□□□□□

□□□□□□□□□□□□□□□□□□□□□□□□□□□□

2026　10

			2026 10

1

Agent Agent

Agent

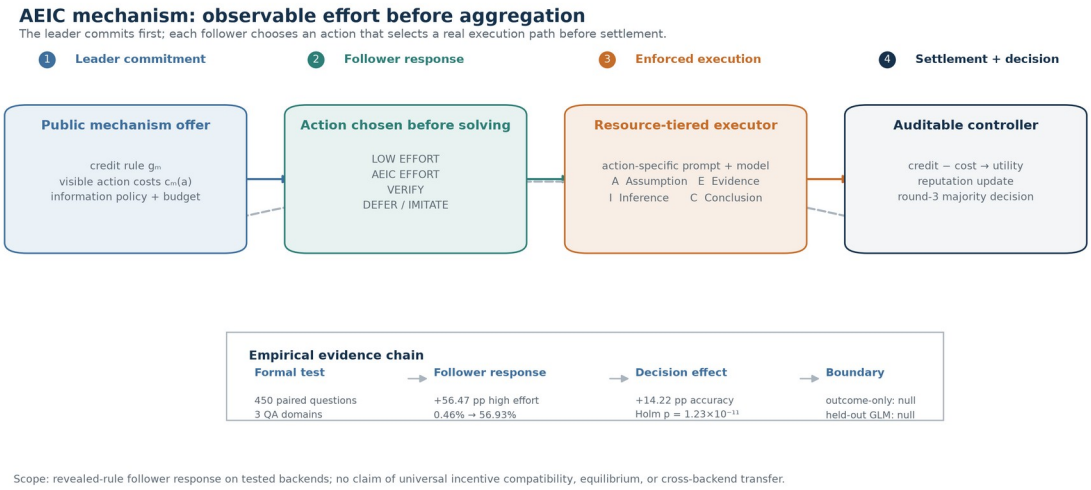
AEIC

Agent Top-K Trace

2

- Prompt Token
- Agent
-

3



1

AEIC Agent r=(A,E,I,C) m=(g,c,P,B)

Low AEIC Verify Imitate

Agent

MovieLens ReDial Preference Diversity

Critic Top-K Trace

4

	AEIC	
	Agent	
	MovieLens	
	JSONL SHA-256	

5

	AEIC Trace	
		Follower
	ReDial	

	AEIC Trace	
	450	
	MovieLens Top- K Web/API	
	1,203 ReDial 100 Agent	

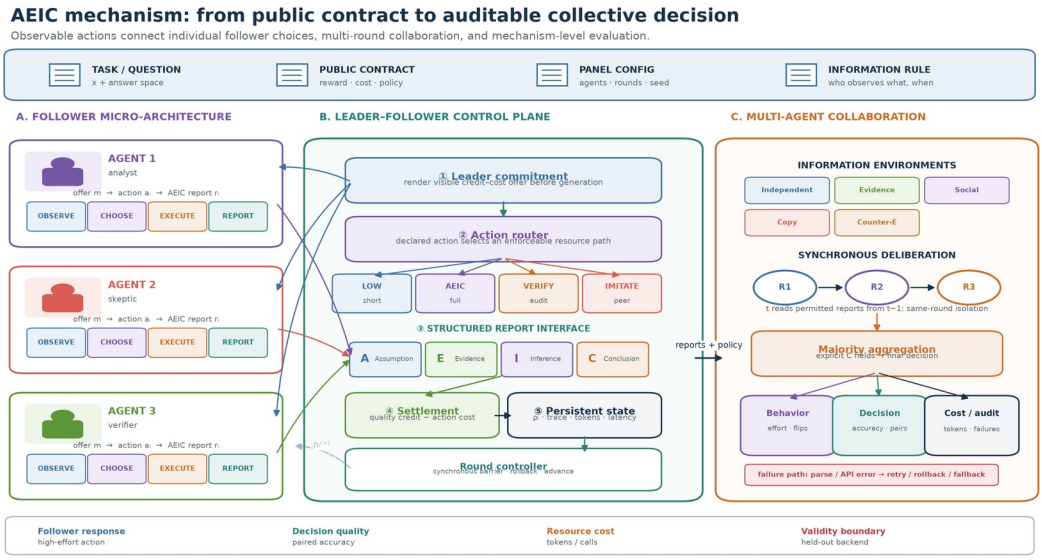
“ ” “

”

1

Leader Follower Agent

Agent P2P



2

complete_aeic consistency novel_evidence verified_evidence useful_correction unsupported_flip Agent

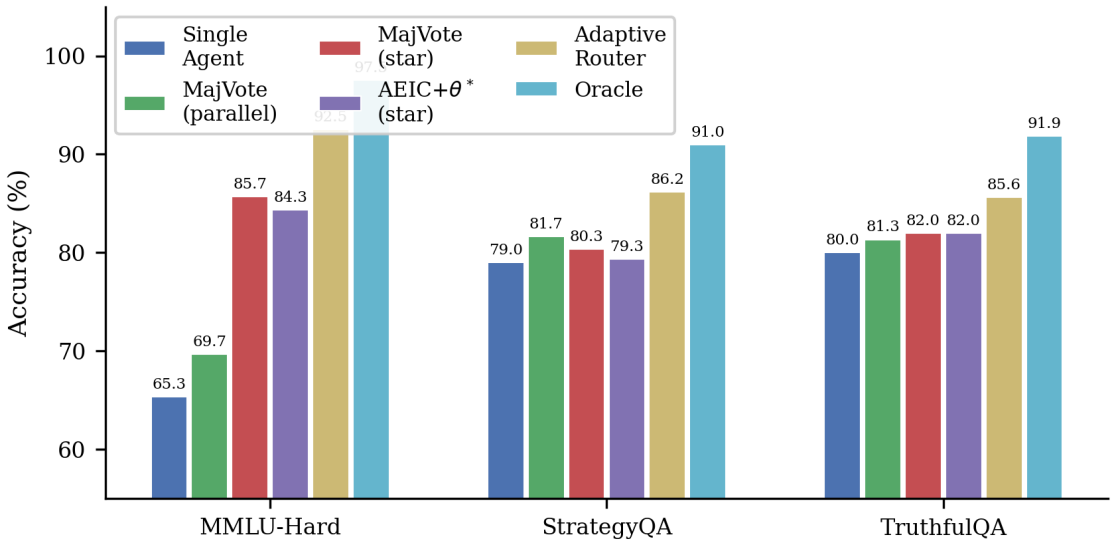
2

StrategyQA MMLU-Hard TruthfulQA 150 450 Agent

No Mechanism Outcome Only Static AEIC

No Mechanism	0.46%	68.00%	
Outcome Only	3.08%	67.33%	−0.67pp
Static AEIC	56.93%	82.22%	+14.22pp

1



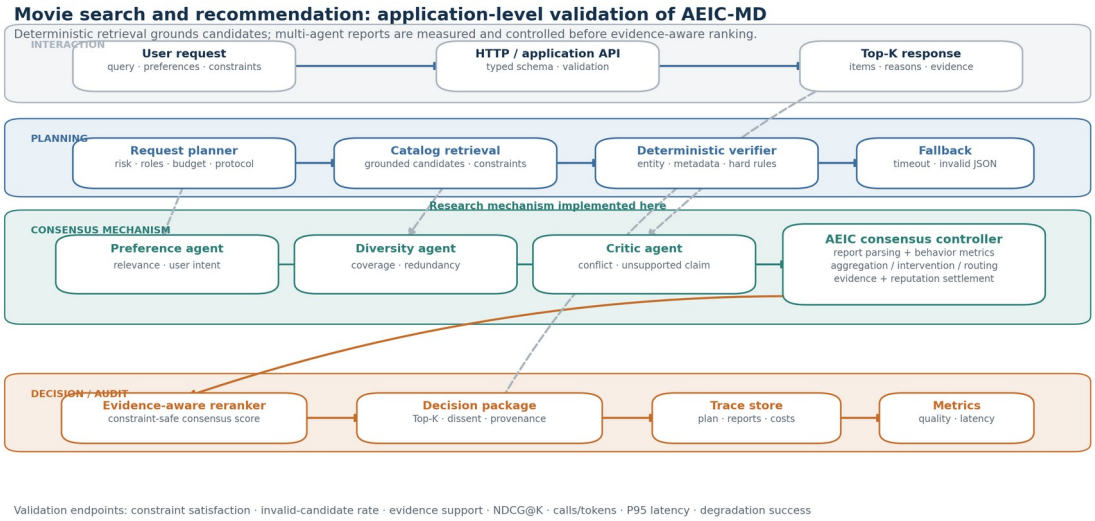
3

Static AEIC 78 14 Holm McNemar p=1.23×10^-11 Agent-round 145.5 Token Outcome Only

AEIC

GLM AEIC

3



4

18,430 MovieLens ReDial Preference Agent Diversity Agent Critic Agent Agent

Trace request_id AEIC Token Agent

4 ReDial

ReDial 8,004 1,342 MovieLens 7,348 1,203 ID

	Hit@5	Recall@5	NDCG@5
	0.75%	0.26%	0.24%
	10.47%	5.83%	4.88%
	10.81%	5.95%	5.01%

2 1,203

10.06 Bootstrap 95% [8.40, 11.89] 123 2 McNemar p=3.70×10^-34

	Hit@5	Recall@5	NDCG@5	
	6%	4.17%	2.66%	
Agent	7%	5.17%	3.11%	5,842 Token / 7.44
Agent	7%	5.17%	2.96%	16,935 Token / 12.68

3 100 Agent

Agent Agent McNemar p=1.0 100% 51.2%

5

	AEIC	
	450	
	API Web Trace	
	MovieLens ReDial	

1

AEIC

2

2026.10-2026.11		
2026.11-2026.12		
2026.12-2027.01		
2027.01-2027.02		
2027.02-2027.03		
2027.04-2027.06		

1

	Agent Agent	
AEIC		
	Agent Token	

2

Follower

Agent Token

P95

[1] Wang L, et al. A survey on large language model based autonomous agents. Frontiers of Computer Science, 2024.

[2] Du Y, et al. Improving factuality and reasoning in language models through multiagent debate. ICML, 2024.

[3] Wei J, et al. Chain-of-thought prompting elicits reasoning in large language models. NeurIPS, 2022.

[4] Yao S, et al. ReAct: Synergizing reasoning and acting in language models. ICLR, 2023.

[5] Shoham Y, Leyton-Brown K. Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations. Cambridge University Press, 2009.

[6] Fudenberg D, Tirole J. Game Theory. MIT Press, 1991.

[7] Koren Y, Bell R, Volinsky C. Matrix factorization techniques for recommender systems. Computer, 2009.

[8] Li R, et al. Towards deep conversational recommendations. NeurIPS, 2018.

[9] Harper F M, Konstan J A. The MovieLens datasets: History and context. ACM TiiS, 2015.

[10] Dietterich T G. Approximate statistical tests for comparing supervised classification learning algorithms. Neural Computation, 1998.

[11] Holm S. A simple sequentially rejective multiple test procedure. Scandinavian Journal of Statistics, 1979.

[12] McNemar Q. Note on the sampling error of the difference between correlated proportions or percentages. Psychometrika, 1947.

BibTeX DOI